Supplementary Material

NeuroBloom: An Open-Source Platform for Longitudinal Cognitive Monitoring and Clinician-Supervised Rehabilitation in Multiple Sclerosis

Table S1: Overview of the 35 NeuroBloom cognitive tasks, including their inspired paradigms, recorded outcomes, and difficulty progression.

Task	Based on	Main results recorded	How difficulty changes
Working memory			
N-Back	N-back task [1]	accuracy, reaction time, missed and false responses	Baseline: 20 trials at 1-back. In training: 1-, 2-, or 3-back depending on level.
Digit Span	Digit span [2]	accuracy, longest sequence, score	10 levels: 3 to 9 digits; recalled forwards, backwards, or in mixed order. Trials 4 and 7 are one level harder.
Spatial Span	Corsi block-tapping [3]	accuracy, longest sequence, score	10 levels: longer sequences and faster display. Difficulty increases when session score is at least 82 and decreases when session score is below 50; otherwise it is unchanged.
Letter–Number Sequencing	Letter–number sequencing [2]	accuracy of items and order, speed, score	10 levels: longer and more mixed sequences with faster presentation.
Operation Span	Operation span [4]	arithmetic accuracy, letter accuracy, score	10 levels: larger sets and more difficult arithmetic operations.
Dual N-Back†	Dual n-back [5]	position accuracy, letter accuracy, combined accuracy, score	10 levels: 1- to 3-back, longer streams, and faster presentation.
Processing speed			
Simple Reaction Time	Simple reaction time [6]	average reaction time, variability, choice accuracy, score	10 levels: changes waiting time and number of response choices.
Symbol–Digit Matching	Symbol Digit Modalities Test [7]	correct and incorrect matches, completion time	10 levels: more symbols and more items.
Numeric Trail	Trail Making Test, part A [8]	completion time, errors, score	10 levels: more nodes and more difficult layouts.
Pattern Comparison	Pattern comparison [9]	accuracy, reaction time, timeouts, score	10 levels: larger and more similar patterns.
Inspection Time	Inspection time [10]	accuracy, shortest correctly identified display time, score	10 levels: shorter presentation time and smaller differences between lines.
Choice Reaction Time†	Choice reaction time [11]	accuracy, reaction time, consistency, score	10 levels: 2 to 4 choices and shorter response windows. Level increases if score is at least 83 and accuracy at least 82%; it decreases if score is below 63.

Task	Based on	Main results recorded	How difficulty changes
Attention			
Continuous Performance	Continuous performance test [12]	hit rate, false responses, reaction time, score	Baseline: 60 trials. In training, 10 trials are added for each level above 3.
Serial Addition	Paced serial addition (PASAT) [13]	correct answers, omissions, reaction time	10 levels: shorter intervals between presented digits.
Color–Word Interference	Stroop test [14]	accuracy, interference effect, reaction time	10 levels: more colors, more incongruent trials, and shorter response time.
Go/No-Go	Go/no-go task [30, 31]	accuracy on go and no-go trials, reaction time, consistency	10 levels: greater stimulus similarity, more no-go trials, and shorter response time.
Flanker	Flanker task [15]	accuracy, interference effect, reaction time	10 levels: more incongruent trials and reduced spacing between stimuli.
Sustained-Attention Response†	Sustained attention to response task [16]	accuracy, commission errors, reaction time	10 levels: longer streams and faster presentation. Level increases if score is at least 84 with at most 4 commission errors; it decreases if score is below 64.
Flexibility			
Task Switching	Cued task switching [17]	accuracy, switch cost, perseverative errors	Baseline: 40 trials. In training, 10 trials are added for each level above 4.
Alternating Trail	Trail Making Test, part B [8]	completion time, errors, difference from part A	10 levels: more nodes and more difficult layouts.
Card-Sorting Rule Shift	Wisconsin Card Sorting Test [18]	categories completed, perseverative errors, accuracy	10 levels: more trials and more frequent rule changes.
Dimensional Card Sort	Dimensional Change Card Sort [19]	accuracy, reaction time, switch cost	10 levels: more rule changes and shorter response time.
Rule Shift†	Cued set shifting [17]	overall accuracy, switch accuracy, reaction time, perseverative errors	10 levels: more rules and faster cues. Level increases if score is at least 83, switch accuracy at least 72%, and perseverative errors at most 5; it decreases if score is below 63.
Plus–Minus Switching	Jersild switching task [20]	accuracy, reaction time, switch cost	10 levels: larger numbers and more mixed blocks.
Planning			
Tower Planning	Tower of London [21]	problems solved, additional moves, time, score	10 levels: more problems and more difficult configurations.
Stockings-Style Planning	Stockings of Cambridge [22]	problems solved, additional moves, time, score	10 levels: more balls and more difficult problems.

Task	Based on	Main results recorded	How difficulty changes
Letter Fluency	Phonemic fluency [23]	number of valid unique words, errors	10 levels: different letter sets and increasing prompt requirements.
Category Fluency†	Semantic fluency [23]	number of valid unique words, repeated words	10 levels: different categories with increasing expected word counts.
Twenty Questions†	Twenty Questions [24]	successful identification, questions used, strategy score	10 levels: different object sets with increasing difficulty.
Visual scanning			
Visual Search†	Visual search [25]	targets found, accuracy, time per target	10 levels: more items and greater similarity between targets and distractors.
Cancellation†	Cancellation test [26]	hits, omissions, incorrect selections, time	10 levels: grids of 150 to 540 cells with increasing target requirements.
Feature and Conjunction Search	Feature-integration search [25]	accuracy, search time	10 levels: larger sets and more similar distractors.
Landmark Line Judgment†	Line bisection [27]	accuracy, omissions, spatial bias, reaction time	10 levels: smaller offsets and shorter presentation. Level increases if score is at least 84, absolute bias is at most 8, and omissions are at most 1; it decreases if score is below 64.
Multiple-Object Tracking†	Multiple-object tracking [28]	accuracy, correct and incorrect selections	10 levels: 6 to 16 objects, 1 to 3 targets, and increasing movement speed.
Central–Peripheral Attention	Useful field of view [29]	central-target and peripheral-target accuracy	Levels 1–3 use a central target only; levels 4–6 add a peripheral target; levels 7–10 add distractors. Presentation time decreases from 800 to 150 ms.

Note. All 35 tasks have automated software-level scoring tests. Tasks marked † also have tests covering additional task behaviour such as trial generation or difficulty adjustment. These tests verify implementation correctness but do not establish psychometric or clinical validity. The psychometric properties of the NeuroBloom task implementations have not yet been established. The cited paradigms serve as conceptual inspirations; trial structure, stimuli, scoring, and difficulty progression are NeuroBloom-specific implementations.

Table S2: Current session-planning and difficulty-adaptation settings used in Algorithm 1.

Parameter	Setting
Initial difficulty	Assigned once when a new training plan is created from the domain baseline score: < 50, level 2; 50–69, level 4; 70–84, level 7; ≥ 85, level 9.
Generic submission pathway: difficulty increase	Increased by one level, up to level 10, when task accuracy in a completed session is at least 85%.
Generic submission pathway: difficulty decrease	Decreased by one level, down to level 1, when task accuracy is below 65%. The level remains unchanged for accuracy from 65% to below 85%.
Focus rebalancing	Reconsidered after every fourth completed session. Domains are ranked using their mean scores from the 12 most recent task results. The two lowest-performing domains become primary focus areas and the next two become secondary focus areas.
Tasks per session	Four tasks are selected: two from primary-focus domains and two from secondary-focus domains.

Note. These settings define the current engineering design of the adaptive planner. They provide a reproducible implementation of baseline-based initialization, difficulty adjustment, and focus rebalancing, and may be refined through future empirical evaluation. Four tasks per session is a developer-selected design choice that allows two primary-focus and two secondary-focus domains to be represented in each session; it has not been empirically optimized.

Table S3: Sensitivity of the longitudinal indicators to selected analytics parameters using synthetic data.

Parameter	Default	Alternatives tested	Effect
Fatigue-proxy weights (accuracy, RT, slope)	0.5, 0.3, 0.2	Equal weighting; accuracy-weighted; RT-weighted	The proxy changed by at most 0.03, 0.10, and 0.15, respectively, on a 0–1 scale. The ordering of the synthetic sessions was unchanged.
Fatigue-proxy normalization constants	30 percentage points, 200 ms, 0.01 per trial	All three constants scaled together by ±20%	The proxy changed by at most 0.04 (−20%) and 0.05 (+20%). The ordering of the synthetic sessions was unchanged.
EWMA smoothing constant α	0.2	0.1, 0.3, 0.5	The trend label changed in 2, 1, and 1 of the 5 synthetic series, respectively.
Trend window	5 points	3 and 7 points	The trend label changed in 1 of 5 series with a three-point window and in 0 of 5 series with a seven-point window.
Trend classification threshold	±1% relative slope	±0.5% and ±2%	The trend label changed in 0 and 3 of the 5 series, respectively. Gradual trends near the decision boundary were most affected.

Parameter	Default	Alternatives tested	Effect
Minimum data requirements	8 valid trials for the fatigue proxy; 3 valid scores for WPSC	Not varied	Increasing these requirements would produce fewer indicator values, whereas reducing them would allow estimates from less data. The current implementation reports insufficient data when the minimum requirements are not met.
Review-flag thresholds	14-day review-flag score: baseline < 45 (+2); recent mean 6–12 points below baseline (+1) or ≥ 12 below (+2); recent mean < 50 (+1); fatigue ≥ 6.5 (+1) or ≥ 8 (+2); completion $< 60\%$ with ≥ 3 sessions (+1). Total ≥ 4 high-priority, 2–3 moderate-priority review flag.	Not varied	Lower cut-offs would assign more patients a moderate- or high-priority review flag. The thresholds are fixed during normal use but can be modified by developers or deployers. They were not varied in the present sensitivity analysis.

Note. Each parameter group was varied separately around its default value on synthetic data (five 40-trial sessions for the fatigue proxy; five 20-session score series for the trend). Minimum data requirements and review-flag thresholds were not varied. This analysis examines how parameter choices affect the generated descriptive indicators and is not intended to identify clinically optimal settings. The displayed proxy ranges (0.3 and 0.5) and CV ranges (0.25 and 0.35) are developer-selected display cut-offs, not clinical thresholds. Missing indicators are not assigned to a range.

Software verification and testing

NeuroBloom is tested with two automated suites. The backend suite contains 118 pytest tests: 31 numerical scoring fixtures, one for each backend-scored task; 43 behavioural tests of trial generation, response classification, and difficulty adjustment; 6 checks of the 35-task inventory; 8 checks of the analytics parameters and sensitivity results; 15 API tests of authentication and clinician endpoints; one integration test of the patient–clinician workflow; 2 tests of focus rebalancing after every fourth completed session; and 12 tests of missing-data handling and analytics correctness. The frontend suite contains 26 Vitest tests covering scoring functions, task navigation, the baseline workflow, progress summaries, and analytics display behaviour.

The integration test exercises the complete workflow through the application programming interface: patient registration and login; clinician registration, administrator approval, and login; the patient's request to a clinician and its approval; submission of the six baseline tasks, baseline calculation, and training-plan generation; completion of a four-task training session, including difficulty adaptation; and retrieval of the patient's overview, session history, and trends by the clinician. It also confirms that a clinician without an approved assignment cannot access the patient's record.

Both suites run on every push and pull request through GitHub Actions (Ubuntu, Python 3.12, Node.js 20). The backend tests use an in-memory SQLite database in place of PostgreSQL. At the time of revision, all 144 tests passed (Windows 10, Python 3.13, Node.js 22). Overall line coverage was approximately 40% for the backend application package and 13% for the frontend source; coverage is higher for the core scoring and analytics logic than for the API, page, and user-interface layers. The frontend tests target scoring and workflow modules; the Svelte page components are not unit-tested, and browser-level end-to-end tests are not yet included.

Reproducibility procedure

The manuscript demonstration can be reproduced from the repository with Docker Compose; the containers use Python 3.12, Node.js 20, and PostgreSQL 16. After the provided environment template (`.env.example`) is copied to `.env.local`, the PostgreSQL database, FastAPI backend, and Svelte frontend are built and started with `compose.yaml`. The database and cognitive-task library are initialized with `seed_initial_data.py`, and `scripts/seed_demo_clinical_data.py` generates a synthetic demonstration dataset of eight clinicians and eight patients with baseline assessments, training plans, repeated training sessions with contextual records, progress reports, and review flags. The script uses a fixed random seed to reproduce the same synthetic patient profiles and performance values; record dates are generated relative to the execution date, so dashboard contents that depend on recent activity differ between runs.

The patient–clinician workflow described in the manuscript can then be followed in the application at `http://localhost:8080` with the demonstration accounts: the patient completes baseline and training tasks, and the clinician reviews the baseline, session history, trends, and longitudinal analytics. Representative analytics outputs are regenerated with `scripts/fatigue_proxy_sensitivity.py` and `scripts/trend_sensitivity.py`, which reproduce the sensitivity results in Table S3, and `scripts/illustrative_case.py`, which reproduces the six-session synthetic case reported in the manuscript. Each script writes a JSON file to `analysis/results/`. The complete commands are given in the repository README. The demonstration accounts are intended for local use only.

Tested environment: Windows 10 Pro (build 19045) with Docker 29.2.0 (Docker Desktop) and Docker Compose v5.0.2; the containers used Python 3.12, Node.js 20, and PostgreSQL 16. The application was accessed using Google Chrome 151.0.7922.138 and Microsoft Edge 154.0.4258.37.

Archived release: NeuroBloom v1.2, `https://doi.org/10.5281/zenodo.23106598`.

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